

MATH1560J Assignment 5 - Sample Answer

Exercise 5.1 (50 pts)

(i)

$$\text{Si}'(x) = \frac{\sin x}{x}, \quad \text{Si}'(0) = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Hence Si increases on intervals where $\sin x > 0$ and decreases where $\sin x < 0$. It oscillates and approaches a finite limit as $x \rightarrow \infty$ (see (iv)). (Sketch: starts at 0, initially increasing, crosses slope zero at $x = \pi$, etc.)

(ii)

Critical points satisfy $\text{Si}'(x) = 0 \iff \sin x = 0$ (for $x \neq 0$), so $x = k\pi$, $k \in \mathbb{Z} \setminus \{0\}$. On $(0, \pi)$, $\sin x > 0$ so Si increases to $x = \pi$; on $(\pi, 2\pi)$ it decreases, so $x = \pi$ is a local maximum. By the same alternation, *local maxima occur at*

$$x = (2k+1)\pi, \quad k = 0, 1, 2, \dots \quad \text{and} \quad x = -2k\pi, \quad k = 1, 2, 3, \dots$$

(iii)

Compute second derivative:

$$\text{Si}''(x) = \frac{d}{dx} \left(\frac{\sin x}{x} \right) = \frac{x \cos x - \sin x}{x^2}.$$

Inflection points occur where $\text{Si}''(x) = 0$, i.e.

$$x \cos x - \sin x = 0 \iff \tan x = x.$$

The first positive nonzero solution of $\tan x = x$ is

$$x_1 \approx 4.4934094579.$$

Thus the first inflection point to the right of the origin is

$$(x_1, \text{Si}(x_1)) \approx (4.4934094579, 1.6555720914).$$

(iv)

$$\lim_{x \rightarrow \infty} \text{Si}(x) = \lim_{x \rightarrow \infty} \int_0^x \frac{\sin t}{t} dt = \int_0^1 \frac{\sin t}{t} dt + \lim_{x \rightarrow \infty} \int_1^x \frac{\sin t}{t} dt$$

$\int_0^1 \frac{\sin t}{t} dt$ obviously exists, so we focus on

$$\lim_{x \rightarrow \infty} \int_1^x \frac{\sin t}{t} dt$$

Apply **Integration by Parts** to $I_X = \int_1^X \frac{\sin t}{t} dt$. Choose $u = \frac{1}{t}$ and $dv = \sin t dt$, and the integration yields:

$$I_X = \left[-\frac{\cos t}{t} \right]_1^X - \int_1^X (-\cos t) \left(-\frac{1}{t^2} \right) dt = \cos 1 - \frac{\cos X}{X} - \int_1^X \frac{\cos t}{t^2} dt$$

By the **Squeeze Theorem**,

$$\lim_{X \rightarrow \infty} \frac{\cos X}{X} = 0 \quad \left(-\frac{1}{X} \leq \frac{\cos X}{X} \leq \frac{1}{X} \right)$$

$$\int_1^\infty \left| \frac{\cos t}{t^2} \right| dt \leq \int_1^\infty \frac{1}{t^2} dt = 1$$

so $\int_1^\infty \left| \frac{\cos t}{t^2} \right|$ converges. Assume it converges to L . Substituting the limits back, there is

$$\int_1^\infty \frac{\sin t}{t} dt = \cos 1 - 0 - L$$

The limit $\lim_{x \rightarrow \infty} \text{Si}(x)$ exists, which proves that $\text{Si}(x)$ has a horizontal asymptote.

(v)

Numerical solution (smallest positive x satisfying $\text{Si}(x) = 1$) is

$$x \approx 1.0648397255 \quad \Rightarrow \quad x \approx 1.1 \text{ (one decimal place).}$$

Exercise 5.2 (20 pts)

We interpret each sum as $\frac{1}{n} \sum_{i=1}^n f\left(\frac{i}{n}\right)$.

(a)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \left(\frac{i}{n} \right)^4 = \int_0^1 x^4 dx = \left[\frac{x^5}{5} \right]_0^1 = \frac{1}{5}.$$

(b)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \sqrt{\frac{i}{n}} = \int_0^1 \sqrt{x} dx = \left[\frac{2}{3} x^{3/2} \right]_0^1 = \frac{2}{3}.$$

Exercise 5.3 (10 pts)

Given $F(x) = \int_{g(x)}^{h(x)} f(t) dt$ with f continuous, g, h differentiable, then by the Fundamental Theorem and chain rule

$$F'(x) = f(h(x)) h'(x) - f(g(x)) g'(x).$$

(Justification: $F(x) = G(h(x)) - G(g(x))$ where $G' = f$, then differentiate.)

Exercise 5.4 (80 pts)

We present substitutions and computations.

(a) $\int e^{\tan x} \sec^2 x \, dx$. Let $u = \tan x$, $du = \sec^2 x \, dx$:

$$\int e^{\tan x} \sec^2 x \, dx = \int e^u \, du = e^{\tan x} + C.$$

(b) $\int \frac{\sin(\ln x)}{x} \, dx$. Let $u = \ln x$, $du = dx/x$:

$$\int \sin(\ln x) \frac{dx}{x} = \int \sin u \, du = -\cos u + C = -\cos(\ln x) + C.$$

(c) $\int \sqrt{\cot x} \csc^2 x \, dx$. Let $u = \cot x$, $du = -\csc^2 x \, dx$:

$$\int \sqrt{\cot x} \csc^2 x \, dx = -\int u^{1/2} \, du = -\frac{2}{3} u^{3/2} + C = -\frac{2}{3} \cot^{3/2} x + C.$$

(d) $\int \frac{dx}{\sqrt{1-x^2} \arcsin x}$. Let $u = \arcsin x$, $du = \frac{dx}{\sqrt{1-x^2}}$:

$$\int \frac{dx}{\sqrt{1-x^2} \arcsin x} = \int \frac{du}{u} = \ln |u| + C = \ln |\arcsin x| + C.$$

(e) $\int_1^2 \frac{e^{1/x}}{x^2} \, dx$. Let $u = \frac{1}{x}$, $du = -\frac{1}{x^2} \, dx$; $x = 1 \mapsto u = 1$, $x = 2 \mapsto u = \frac{1}{2}$.

$$\int_1^2 \frac{e^{1/x}}{x^2} \, dx = \int_{u=1}^{1/2} -e^u \, du = \int_{1/2}^1 e^u \, du = e^1 - e^{1/2}.$$

(f) $\int_{-\pi/3}^{\pi/3} x^4 \sin x \, dx$. Note x^4 is even, $\sin x$ is odd $\Rightarrow$ integrand is odd. Over symmetric interval:

$$\int_{-\pi/3}^{\pi/3} x^4 \sin x \, dx = 0.$$

(g) $\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}}$. Let $u = \ln x$, $du = \frac{dx}{x}$; $u \in [1, 4]$:

$$\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}} = \int_1^4 u^{-1/2} \, du = \left[2u^{1/2} \right]_1^4 = 2(2 - 1) = 2.$$

(h) $\int_0^1 \frac{dx}{(1+\sqrt{x})^4}$. Let $u = \sqrt{x}$, $dx = 2u \, du$, $u \in [0, 1]$:

$$\int_0^1 \frac{dx}{(1+\sqrt{x})^4} = \int_0^1 \frac{2u}{(1+u)^4} \, du.$$

Set $t = 1 + u$ ($t \in [1, 2]$): integrand $2(t-1)/t^4 = 2(t^{-3} - t^{-4})$.

$$\int_1^2 2(t^{-3} - t^{-4}) \, dt = \left[-t^{-2} + \frac{2}{3}t^{-3} \right]_1^2 = \left(-\frac{1}{4} + \frac{1}{12} \right) - \left(-1 + \frac{2}{3} \right) = \frac{1}{6}.$$

Exercise 5.5 (20 pts)

(a) $I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx.$

Use symmetry: substitute $x \mapsto \pi - x$ gives

$$I = \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx.$$

Adding the two expressions:

$$2I = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx.$$

Let $u = \cos x$, $du = -\sin x dx$; $u : 1 \rightarrow -1$:

$$\int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx = \int_1^{-1} \frac{-du}{1 + u^2} = \int_{-1}^1 \frac{du}{1 + u^2} = [\arctan u]_{-1}^1 = \frac{\pi}{2}.$$

Hence $2I = \pi \cdot \frac{\pi}{2}$, so

$$I = \frac{\pi^2}{4}.$$

(b) $\int_0^{\pi/2} \frac{dx}{1 + \tan^{2025} x}.$

For n odd, use $x \mapsto \frac{\pi}{2} - x$:

$$\int_0^{\pi/2} \frac{dx}{1 + \tan^n x} = \int_0^{\pi/2} \frac{dx}{1 + \cot^n x} = \int_0^{\pi/2} \frac{\tan^n x}{1 + \tan^n x} dx.$$

Adding both representations yields $2I = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$. Thus

$$I = \frac{\pi}{4}.$$

(So here $I = \pi/4$ for $n = 2025$ odd.)

Exercise 5.6 (30 pts)

(i)

Define $F(u) = \int_0^u \frac{1}{\sqrt{1+t^3}} dt$ so $F'(u) = \frac{1}{\sqrt{1+u^3}}$. By chain rule

$$f'(x) = F'(g(x)) g'(x) = \frac{1}{\sqrt{1+g(x)^3}} g'(x).$$

Compute $g'(x)$. Let $H(u) = \int_0^u (1 + \sin(t^2)) dt$, so $g(x) = H(\cos x)$ and $H'(u) = 1 + \sin(u^2)$. Thus

$$g'(x) = H'(\cos x) \cdot (-\sin x) = (1 + \sin(\cos^2 x))(-\sin x).$$

Evaluate at $x = \pi/2$: $\cos(\pi/2) = 0$, hence $g(\pi/2) = 0$, $\sin(\cos^2(\pi/2)) = 0$, $\sin(\pi/2) = 1$. Therefore

$$f'(\frac{\pi}{2}) = \frac{1}{\sqrt{1+0^3}} \cdot (1+0) \cdot (-1) = -1.$$

$f'(\pi/2) = -1.$

(ii)

We have:

$$f(x) = \int_0^x x^2 \sin(t^2) dt = x^2 \int_0^x \sin(t^2) dt$$

Differentiate using the product rule:

$$f'(x) = \frac{d}{dx} \left[x^2 \int_0^x \sin(t^2) dt \right] = 2x \int_0^x \sin(t^2) dt + x^2 \cdot \frac{d}{dx} \left[\int_0^x \sin(t^2) dt \right]$$

By the Fundamental Theorem of Calculus:

$$\frac{d}{dx} \left[\int_0^x \sin(t^2) dt \right] = \sin(x^2)$$

Therefore:

$$f'(x) = 2x \int_0^x \sin(t^2) dt + x^2 \sin(x^2)$$

$$2x \int_0^x \sin(t^2) dt + x^2 \sin(x^2)$$

(iii)

Evaluate:

$$\lim_{x \rightarrow 0} \frac{1}{x} \int_0^x (1 - \tan 2t)^{1/t} dt$$

Let:

$$F(x) = \int_0^x (1 - \tan 2t)^{1/t} dt$$

Then the limit becomes:

$$\lim_{x \rightarrow 0} \frac{F(x)}{x}$$

This is a 0/0 indeterminate form. Apply L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{F(x)}{x} = \lim_{x \rightarrow 0} F'(x)$$

By the Fundamental Theorem of Calculus:

$$F'(x) = (1 - \tan 2x)^{1/x}$$

Therefore:

$$\lim_{x \rightarrow 0} \frac{F(x)}{x} = \lim_{x \rightarrow 0} (1 - \tan 2x)^{1/x}$$

Now analyze the limit:

$$\lim_{x \rightarrow 0} (1 - \tan 2x)^{1/x}$$

Take natural logarithm:

$$\ln L = \lim_{x \rightarrow 0} \frac{\ln(1 - \tan 2x)}{x}$$

As $x \rightarrow 0$, $\tan 2x \approx 2x$, so:

$$\ln(1 - \tan 2x) \approx \ln(1 - 2x) \approx -2x$$

Thus:

$$\ln L = \lim_{x \rightarrow 0} \frac{-2x}{x} = -2$$

Therefore:

$$L = e^{-2}$$

$$\boxed{e^{-2}}$$

Exercise 5.7 (10 pts)

Substitute $u = \frac{ab}{x}$, so $x = \frac{ab}{u}$ and $dx = -\frac{ab}{u^2} du$. When $x = a$, $u = b$; when $x = b$, $u = a$. Thus

$$\int_a^b f(x) dx = \int_{u=b}^{u=a} f\left(\frac{ab}{u}\right) \left(-\frac{ab}{u^2}\right) du = \int_a^b \frac{ab}{x^2} f\left(\frac{ab}{x}\right) dx,$$

renaming u back to x .

Exercise 5.8 (10 pts)

Proof:

Since g is a C^1 -bijection, either $g'(t) > 0$ for all t or $g'(t) < 0$ for all t .

Case 1: $g'(t) > 0$

Then g is strictly increasing, so:

$$g(\alpha) = a, \quad g(\beta) = b$$

Using the substitution $x = g(t)$, $dx = g'(t)dt$:

$$\int_a^b f(x)dx = \int_{\alpha}^{\beta} f(g(t))g'(t)dt = \int_{\alpha}^{\beta} f(g(t))|g'(t)|dt$$

since $g'(t) > 0$ implies $|g'(t)| = g'(t)$.

Case 2: $g'(t) < 0$

Then g is strictly decreasing, so:

$$g(\alpha) = b, \quad g(\beta) = a$$

Using the substitution $x = g(t)$, $dx = g'(t)dt$:

$$\int_a^b f(x)dx = \int_{\alpha}^{\beta} f(g(t))g'(t)dt$$

Since $g'(t) < 0$, we have:

$$\int_{\alpha}^{\beta} f(g(t))g'(t)dt = - \int_{\alpha}^{\beta} f(g(t))|g'(t)|dt$$

But when g is decreasing, the correct substitution should account for reversed limits:

$$\int_a^b f(x)dx = \int_b^a f(g(t))g'(t)dt = \int_\alpha^\beta f(g(t))(-g'(t))dt = \int_\alpha^\beta f(g(t))|g'(t)|dt$$

where we used $|g'(t)| = -g'(t)$ when $g'(t) < 0$.

In both cases, we obtain:

$$\int_a^b f(x)dx = \int_\alpha^\beta f(g(t))|g'(t)|dt$$

$$\int_a^b f(x)dx = \int_\alpha^\beta f(g(t))|g'(t)|dt$$

Numerical values quoted:

First solution of $\tan x = x$ (first inflection point): $x \approx 4.4934094579$, and $\text{Si}(4.4934094579) \approx 1.6555720914$.

Solution of $\text{Si}(x) = 1$ (smallest positive root): $x \approx 1.0648397255$ (one-decimal: 1.1).

c. **Proof of** $\int_0^\infty \frac{\sin z}{z} dz = \frac{\pi}{2}$

As an application of the lemma we evaluate the integral

$$(19) \quad I = \int_0^\infty \frac{\sin z}{z} dz.$$

This improper integral is defined by the relation

$$I = \lim_{M \rightarrow \infty} I_M$$

where

$$I_M = \int_0^M \frac{\sin z}{z} dz.$$

The convergence of the improper integral I , that is, the existence of the limit of I_M for $M \rightarrow \infty$ had been proved already on p. 310. The convergence proof was based on integration by parts and may be restated here. If, say, $0 < M < N$, we have

$$(20) \quad \begin{aligned} |I_N - I_M| &= \left| \int_M^N \frac{\sin z}{z} dz \right| \\ &= \left| \left. \frac{-\cos z}{z} \right|_M^N + \int_M^N \frac{\cos z}{z^2} dz \right| \\ &\leq \frac{1}{M} + \frac{1}{N} + \int_M^N \frac{dz}{z^2} = \frac{2}{M}. \end{aligned}$$

Since then I_N and I_M differ arbitrarily little if both M and N are sufficiently large, the existence of $I = \lim_{M \rightarrow \infty} I_M$ is assured by Cauchy's convergence test. Moreover, letting N tend to infinity in (20) we find an estimate for the rate at which the I_M approach their limit I :

$$(20a) \quad |I - I_M| \leq \frac{2}{M}.$$

We can rewrite our expression for I in such a way that I appears as a limit of integrals over a fixed finite interval. Let p be an arbitrary positive number; for $M = \lambda p$ the substitution $z = \lambda x$, $dz = \lambda dx$ shows that

$$I_{\lambda p} = \int_0^{\lambda p} \frac{\sin z}{z} dz = \int_0^p \frac{\sin \lambda x}{x} dx.$$

Since $\lambda p \rightarrow \infty$ for $\lambda \rightarrow \infty$ and fixed positive p , we clearly have

$$I = \lim_{\lambda \rightarrow \infty} \int_0^p \frac{\sin \lambda x}{x} dx,$$

and more precisely from (20a)

$$\left| I - \int_0^p \frac{\sin \lambda x}{x} dx \right| < \frac{2}{\lambda p}.$$

Thus for any positive p the expressions

$$\int_0^p \frac{\sin \lambda x}{x} dx$$

approach for $\lambda \rightarrow \infty$ one and the same value I ; moreover, the convergence is uniform in p as long as we restrict p to values above some fixed positive number P . Indeed, the difference between the integral and the limit I is then less than ϵ for $\lambda > 2/P\epsilon$.

We now apply our lemma of p. 588 to the function

$$k(x) = \frac{1}{x} - \frac{1}{2 \sin(x/2)}$$

If we define $k(0) = 0$, the function $k(x)$ is continuous and has a continuous first derivative for $0 \leq x < 2\pi$ (see p. 466). Hence our lemma shows that

$$\int_0^p \sin \lambda x \left(\frac{1}{x} - \frac{1}{2 \sin(x/2)} \right) dx$$

tends to zero for $\lambda \rightarrow \infty$ as long as $0 \leq p < 2\pi$. Moreover, by (18) the convergence is uniform for $0 \leq p \leq \pi$ since $|k(x)|$ and $|k'(x)|$ are bounded in the interval $0 \leq x \leq \pi$. It follows from our previous result that for any p in the interval $0 < p < 2\pi$

$$\lim_{\lambda \rightarrow \infty} \int_0^p \frac{\sin \lambda x}{2 \sin(x/2)} dx = I,$$

and also that the convergence is uniform in p for $P \leq p \leq \pi$, where P is a fixed positive number.

Now for $p = \pi$ and $\lambda = n + \frac{1}{2}$ (where n is an integer) we have evaluated this integral [see formula (15), p. 586] and found that it has the value $\pi/2$ independently of n . Letting λ tend to infinity through values of the form $\lambda = n + \frac{1}{2}$, we find then for I the value $\pi/2$:

$$(21) \quad \int_0^\infty \frac{\sin z}{z} dz = \frac{\pi}{2}.$$